Preprocessing and Normalization Summary Report

Results for SGCC, CER, London LCL and ETD2022

Version 1.1 | Collection CER4 | 27 October 2026

Abstract. All four datasets, SGCC, CER, London LCL and ETD2022, have been cleaned and put into one common form. Each is reduced to a daily record in kilowatt hours with the same columns, the same rules for gaps and outliers, five partitions by meter, the same absolute and relative scales, and windows of 30 days. The final form holds 1,275,674 windows of 30 daily values for each seed, with identical columns for every dataset. The real calendars differ: SGCC and London LCL overlap for 58 days, CER overlaps with neither, and ETD2022, which has no calendar, carries a nominal one. Scaling by the literal minimum and maximum of the SGCC training consumers compresses the median reading of SGCC, CER and London LCL to between 0.000006 and 0.000045 across the five seeds, so a robust scale should be assessed before modelling. The findings that affect modelling are listed in Table 13.

1 Status and Common Basis

Table 1 gives the state of each dataset and Table 2 states what has been made common. The rules behind both tables are in the Preprocessing Protocol and the Normalization Protocol.

Table 1. State of each dataset.

Dataset	Verified and cleaned	In the common daily table	Normalized views	30 day windows	Remark
SGCC	Yes	Yes	Yes	1,060,048	Labels at consumer level
CER	Yes	Yes	Yes	104,101	No theft labels
London LCL	Yes	Yes	Yes	110,757	No theft labels. ACORN attributes kept as covariates
ETD2022	Yes	Yes, on a nominal calendar	Yes	768	Inferred blocks, power in kW, synthetic labels, test only

Table 2. Properties made common across the four datasets.

Property	Common specification
Time resolution	One value per meter and day
Energy unit	Energy in kWh per day. ETD2022 is converted from hourly kW under the stated assumption
Record key	`domain`, `meter_id`, `timestamp`
Calendar basis	A real calendar day for SGCC, CER and London LCL, and a nominal day for ETD2022, marked by `calendar_basis`
Cleaning	One set of rules for duplicates, invalid slots, negative values, short gaps, long gaps and outliers, applied at the native interval before daily summation
Eligibility	Only a complete day carries a kWh value, as recorded in `analysis_eligible`
Partitions	Five seeds, with every meter in exactly one of training, validation and test. ETD2022 is test only
Scales	An absolute scale fitted on SGCC training consumers and a relative profile per meter. Physical kWh is retained
Sequence length	Nonoverlapping windows of 30 consecutive eligible days
Final form	One record of 30 daily values per window, with identical columns for all four datasets, one table per seed

2 Cleaning Outcomes

Table 3 reports the effect of the cleaning rules on each dataset. Raw files are unchanged.

Table 3. Cleaning outcomes for the four datasets.

Measure	SGCC	CER	London LCL	ETD2022
Native interval	Daily	30 minutes	30 minutes	Hourly
Expected readings on the full grid	43,855,020	158,484,746	168,268,119	560,640
Readings present after duplicates and invalid slots	32,579,120	157,992,424	167,811,461	560,640
Readings missing	11,275,900	492,322	456,658	0
Zero readings, kept as valid	5,788,603	1,557,479	2,001,552	22,958
Negative readings	0	0	0	0
Exact duplicate rows removed	0	0	115,453 in the publisher file, absent from the processed copy	Not applicable
Conflicting duplicate slots invalidated	0	12	0	Not applicable
Invalid or off grid rows removed	0	560	5,560	15
Short gaps interpolated, in intervals	326,500	12	19,224	0
Long gap and boundary readings left null	10,949,400	492,322	437,434	0
Readings above five times the 99th percentile, flagged	872	682	1,269	0
Daily rows	43,855,020	3,301,514	3,507,915	23,360
Eligible daily rows	32,905,620	3,291,245	3,491,845	23,360
Share of daily rows eligible	75.0%	99.7%	99.5%	100.0%
Meters without any eligible day	5	0	1	0

Note. For SGCC and London LCL the missing readings equal the interpolated readings plus the long gap and boundary readings. For CER they equal the long gap and boundary readings, and the 12 interpolated intervals all lie on quarantined days, so none enters an eligible daily total. Five further London households appear only in the metadata and have no readings. ETD2022 counts exclude the 15 quarantined source rows, in which every feature is zero. Its zero readings include the Theft2 attacks.

3 Temporal Coverage and Consumption Level

The common unit is the day. It is not a common calendar period. Table 4 shows where each domain lies in time, and Figure 1 shows how many meters contribute on each real calendar day.

Table 4. Coverage and daily consumption of the eligible readings.

Domain	First eligible day	Last eligible day	Days in table	Days shared with another domain	Mean kWh per day	Median kWh per day	99th percentile kWh per day
SGCC	1 Jan 2014	31 Oct 2016	1,035	58	9.35	4.59	69.84
CER	14 Jul 2009	31 Dec 2010	536	0	33.87	22.91	285.54
London LCL	24 Nov 2011	27 Feb 2014	828	58	10.16	7.84	45.84
ETD2022	1 Jan 1971, nominal	31 Dec 1971, nominal	365	0	3,882.72	1,156.83	29,070.11

Note. The maximum daily reading is 800,003 kWh for SGCC, 1,641 kWh for CER, 332 kWh for London LCL and 32,094 kWh for ETD2022. The London LCL table begins on 23 November 2011 with a partial day that is not eligible. The ETD2022 calendar is artificial, and its days are counted per block.

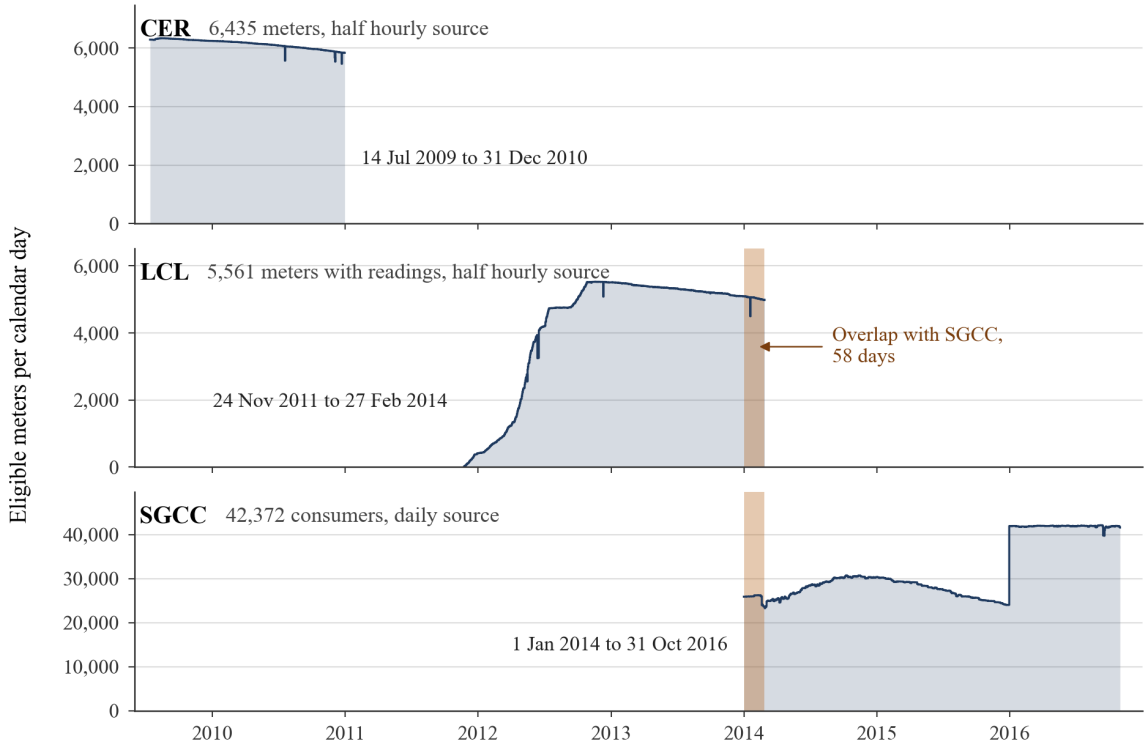


Figure 1. Eligible meters per calendar day in each real domain. The shaded band marks the 58 days, 1 January to 27 February 2014, on which SGCC and London LCL both have eligible readings. CER has no day in common with either. ETD2022 has a nominal calendar and is not shown.

4 Normalization Outcomes

4.1 Partitions

Each of the five seeds assigns every meter to one partition. The counts below are the same for all seeds, and only the membership changes. ETD2022 is the single outside check of the research plan, so all 64 blocks are test.

Table 5. Meters per partition.

Domain	Training	Validation	Test	Theft labelled consumers (training / validation / test)
SGCC	29,659	4,236	8,477	2,530 / 361 / 724
CER	3,861	643	1,931	Not applicable
London LCL	3,336	556	1,669	Not applicable
ETD2022	0	0	64	Not applicable, labels are per day
Total	**36,856**	**5,435**	**12,141**	**3,615**

4.2 Absolute scale and relative profile

Table 6 shows where typical readings fall on the absolute scale of seed 20261021. Table 7 gives the relative profile, which is identical for all seeds because it uses only each meter's own mean.

Table 6. Absolute scaled values by domain, seed 20261021.

Domain	Median kWh per day	Median scaled	99th percentile scaled	Maximum scaled
SGCC	4.59	0.000006	0.000087	1.000000
CER	22.91	0.000029	0.000357	0.002052
London LCL	7.84	0.000010	0.000057	0.000415
ETD2022	1,156.83	0.001446	0.036337	0.040117

Note. With seeds 20261022 and 20261024 the SGCC training maximum is lower, and then 4 and 1 SGCC validation or test readings exceed 1. The largest is 1.553. No value was clipped.

Table 7. Relative profile by domain.

Domain	Undefined values (zero mean meters)	Median	99th percentile	Maximum
SGCC	1,444,044	0.844	4.621	1,035.0
CER	0	0.956	2.542	59.9
London LCL	647	0.965	2.343	409.3
ETD2022	0	1.128	1.921	2.566

4.3 Windows of 30 days

Windows of 30 consecutive eligible days with a stride of 30 days were indexed for every meter. Days that cannot complete a window are logged and left unused.

Table 8. Windows of 30 days by domain.

Domain	Eligible days	Contiguous runs	Windows	Days used	Unused days	Windows with a relative profile
SGCC	32,905,620	88,357	1,060,048	96.6%	1,104,180	1,013,607
CER	3,291,245	7,739	104,101	94.9%	168,215	104,101
London LCL	3,491,845	12,770	110,757	95.2%	169,135	110,737
ETD2022	23,360	64	768	98.6%	320	768
Total	**39,712,070**	**108,930**	**1,275,674**	**96.4%**	**1,441,850**	**1,229,213**

4.4 Final common form

The final form stores every window with the same columns for all four datasets. Table 9 gives the windows of seed 20261021. The label counts do not depend on the seed, but the counts per partition do.

5 ETD2022 Outcomes

ETD2022 was verified and cleaned, and it follows the common record. Fifteen of its 560,655 rows, in which every feature is zero, were quarantined. The 560,640 clean hourly rows come from 16 building archetypes and form 64 inferred blocks. Each block is treated as a meter with 365 days on a nominal calendar. A day is labelled theft when at least one of its hours carries an attack label.

Table 9. Final windows by dataset and partition, seed 20261021.

Domain	Partition	Windows	Label 1	Label 0	Label unknown
SGCC	Training	740,457	57,711	682,746	0
	Validation	105,712	8,243	97,469	0
	Test	213,879	16,668	197,211	0
CER	Training	62,284	0	0	62,284
	Validation	10,439	0	0	10,439
	Test	31,378	0	0	31,378
London LCL	Training	66,478	0	0	66,478
	Validation	11,160	0	0	11,160
	Test	33,119	0	0	33,119
ETD2022	Test	768	238	530	0
Total		**1,275,674**	**82,860**	**977,956**	**214,858**

Note. A window is labelled 1 when at least half of its 30 days carry a theft label. For SGCC this is the consumer flag. For ETD2022 a day carries the label when at least one of its hours is an attack hour. No ETD2022 window is free of attack days, since the share of labelled days ranges from 0.13 to 0.73 with a median of 0.43.

Table 10. Hourly labels in the clean ETD2022 table.

Label	Hours	Share of hours
Normal	331,824	59.2%
Theft1	51,083	9.1%
Theft2	22,958	4.1%
Theft3	44,349	7.9%
Theft4	41,460	7.4%
Theft5	33,553	6.0%
Theft6	35,413	6.3%
All attack labels	**228,816**	**40.8%**
Total	**560,640**	**100.0%**

Table 11. Composition of the 23,360 days.

Type of day	Days	Share of days
All 24 hours Normal	13,139	56.2%
All 24 hours carry the same attack label	8,254	35.3%
Mixed labels	1,967	8.4%
Mixed labels, Normal and attack hours	1,337	5.7%
Mixed labels, several attack types	630	2.7%
Days labelled theft, at least one attack hour	**10,221**	**43.8%**

Note. The first three rows are exclusive and add to 23,360 days. The two rows below Mixed labels are subsets of it. The daily energy assumes that each hourly kW value is the mean power over its hour.

6 Validation Results

Table 12. Validation results.

Check	Result
Source integrity	All checksum and archive tests passed. The three SGCC copies are identical
Row reconciliation	Exact for SGCC, CER and London LCL
ETD2022 daily table	Daily kWh equals the hourly sums, with a largest relative difference of 0.00000006. Day labels equal the presence of an attack hour, and every block has 365 consecutive nominal days
Independent normalization run	Values, partitions, scalers, windows and window identifiers of SGCC, CER and London LCL equal those of an independent normalization run made on 23 October 2026
Daily table	No eligible day has a null value and no daily value is negative
Normalized tables	Five tables of 39,712,070 rows each, equal to the eligible rows of the four datasets
Partitions	272,160 assignments. No meter is in more than one partition for any seed, and every ETD2022 block is test
Absolute scale	The scaled SGCC training range is 0 to 1 for every seed. Recomputation of every value agrees within 0.0000001
Relative profile	The mean per meter equals 1 within 0.0000001. Values are empty exactly for zero mean meters
Windows	All 1,275,674 windows span 30 days with unique identifiers. 3,000 selected windows and all 768 ETD2022 windows are complete against the source days
Final windows	Splits equal the manifest, every array holds 30 values, and labels are as stated. 2,268 windows, which include every ETD2022 window, equal the normalized daily values exactly in each of the five seeds
Excel workbooks	All 66 workbooks equal their source tables in row count and, for every column, in the count of values and the sum, minimum and maximum, the distinct count, or the count of true values, as applicable. The parts of every table add up to its rows. Every workbook has one sheet with a frozen header row, and none exceeds one Excel sheet

7 Findings That Affect Modelling

Table 13. Findings, their evidence and their implication.

Finding	Evidence	Implication
Calendar periods differ	SGCC and London LCL share 58 days, from 1 January to 27 February 2014. CER shares none. ETD2022 days are nominal.	The domains cannot be aligned in time. Transfer between them is a domain shift.
Consumption levels differ	Mean daily kWh is 33.87 for CER, 10.16 for London LCL, 9.35 for SGCC and 3,882.72 for ETD2022.	One absolute scale does not equalize the domains. ETD2022 buildings consume hundreds of times more than SGCC and London LCL customers. CER customer segments are not joined and may contribute to its level.
Literal scaling compresses	Median scaled values are 0.000006 to 0.000045 for SGCC, CER and London LCL across the seeds, and 0.0014 to 0.0022 for ETD2022. The SGCC training maximum is 514,992 to 800,003 kWh per day.	Assess a robust scale fitted on SGCC training consumers only. Physical kWh is retained.
SGCC coverage is uneven	75.0% of daily rows are eligible. Eligible consumers per day number 23,253 to 30,713 in 2014 and 2015, and 39,710 to 42,108 in 2016.	Series lengths differ between consumers. A window exists only where 30 consecutive eligible days exist.
SGCC omits one date	18 September 2016 is absent for every consumer. It was interpolated for 39,710 consumers, 12.2% of the 326,500 interpolated days, and stays ineligible for 2,662.	An interpolated day is not an observation. The column `is_imputed` of the SGCC table and `imputed_days` of the final windows identify it.

Table 13 (continued). Findings, their evidence and their implication.

Finding	Evidence	Implication
Zero mean meters	2,110 meters with eligible days have a zero mean (2,109 SGCC and 1 London LCL), covering 1,444,691 days. Six further meters have no eligible day.	The relative profile is undefined for them, so their windows have empty relative arrays and `relative_defined` is FALSE.
Labels differ in kind and prevalence	3,615 of 42,372 SGCC consumers (8.5%) carry a theft label at consumer level. CER and London LCL have none. ETD2022 labels are synthetic, and 43.8% of its days and 31.0% of its windows are labelled theft.	No dated theft event is available. ETD2022 prevalence differs strongly from SGCC. Synthetic attack injection for CER is a later step.
ETD2022 is test only	The research plan treats it as a single outside check, so all 64 blocks are test in every seed.	Training on ETD2022 would need a new partition, which is a change of the split column only.
ETD2022 identity and calendar are inferred	Blocks are inferred sequences, the calendar is nominal, and energy is estimated from power. Its window label uses the threshold of one half.	Do not use weekday or month features for ETD2022. Use label_fraction for other thresholds and the hours per label of the ETD2022 daily table for evaluation by attack type.
Daylight saving in CER	Days with 46 or 50 slots keep their observed totals.	Consider exclusion or adjustment in a sensitivity analysis.
Departure from the plan	The research plan proposes zero filling of long gaps and a three sigma cap. The supervisor's rule of exclusion and a flag at five times the 99th percentile was followed.	Confirm this choice before modelling.

8 Location of the Datasets

All paths are relative to the collection folder CER4. The CER dataset that was processed is in `1_preprocessing/cer`. Its windows are in the final window tables under the domain value cer. The original CER archives are not delivered. The datasets are delivered as Excel workbooks in three stage folders. A table that fits one Excel sheet is in one workbook, and a larger table is in consecutive numbered workbooks of at most 1,000,000 rows. Every workbook holds complete rows of its table.

Table 14. Folders of the collection.

Content	Folder	Workbooks
SGCC	`1_preprocessing/sgcc`	`sgcc_daily` in 44 parts
CER	`1_preprocessing/cer`	`cer_daily` in 4 parts
London LCL	`1_preprocessing/lcl`	`lcl_daily` in 4 parts
ETD2022	`1_preprocessing/etd2022`	`etd_daily`
Partitions, scalers and meter means	`2_normalization`	`split_manifest_five_seeds`, `scaler_parameters`, `meter_summary`
Final common normalized form	`3_final_common_normalized`	`final_windows_30d_seed_20261021` to `final_windows_30d_seed_20261025`, each in 2 parts
Documents	`Documents`	PDF, Markdown and LaTeX versions of the Protocol, Preprocessing Protocol, Normalization Protocol and Summary Report